

NPTEL Course on "Computer System Optimization for ML"

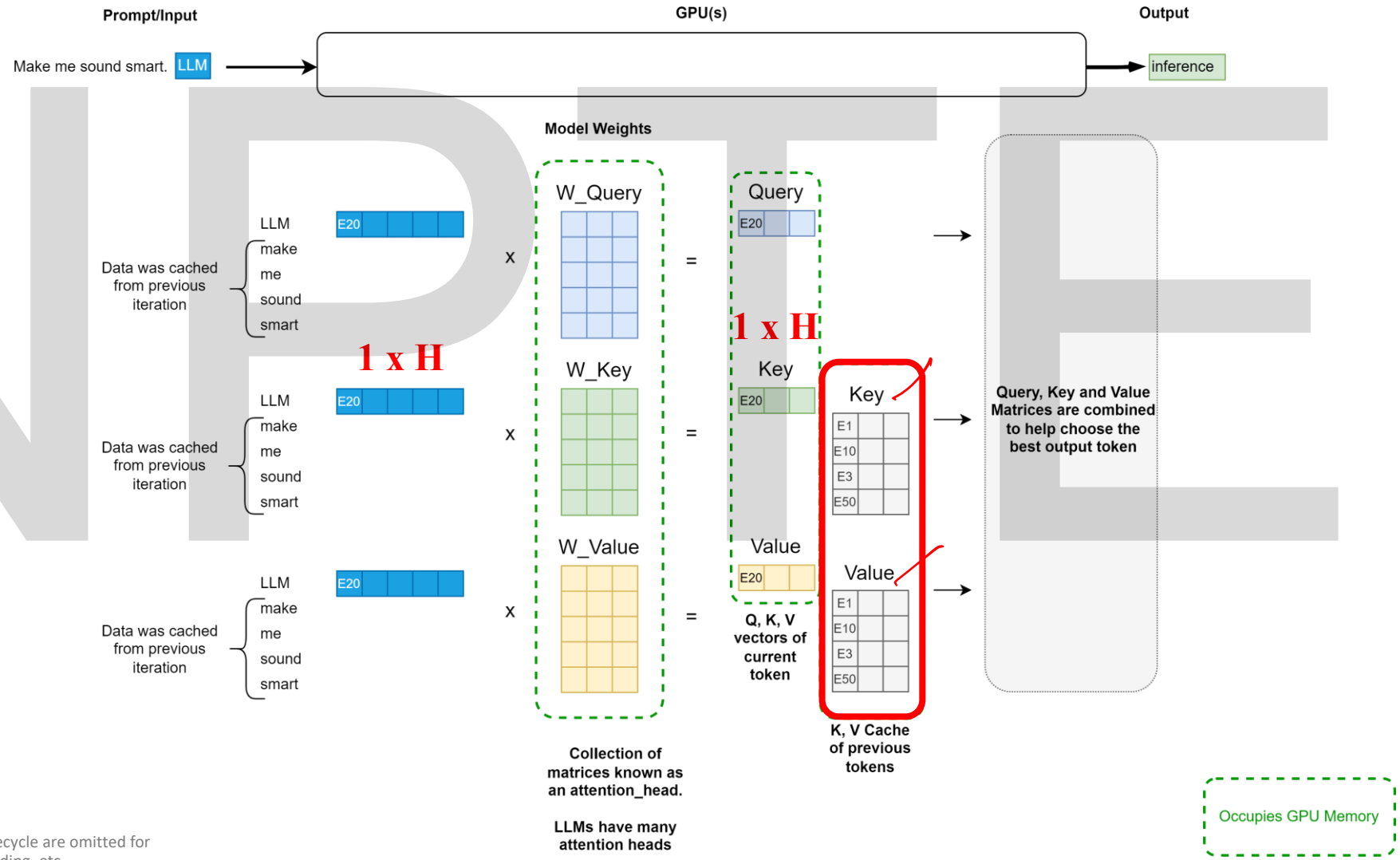
Lecture #28

Agenda

- KV-Cache (contd.)
- Paged-attention

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KV cache to the rescue!



Many details of the model inference lifecycle are omitted for brevity such as mlps, layer-norms, decoding, etc

KV cache memory use

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N_{layers} : layers in the model

d_{model} : dimension of embeddings

$seqlen$: length of context in tokens

KV cache memory use

Size of K, V matrices



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KV cache memory use

Size of K, V matrices

$2 *$ $d_{model} * seqlen$

N_{layers} : layers in the model

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$seqlen$: length of context in tokens

d model x seq-length

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KV cache memory use

Size of K, V matrices

$2 *$ $d_{model} * seqlen$

$precision$	:	bytes per parameter (e.g. 4 for fp32)
N_{layers}	:	layers in the model
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KV cache memory use

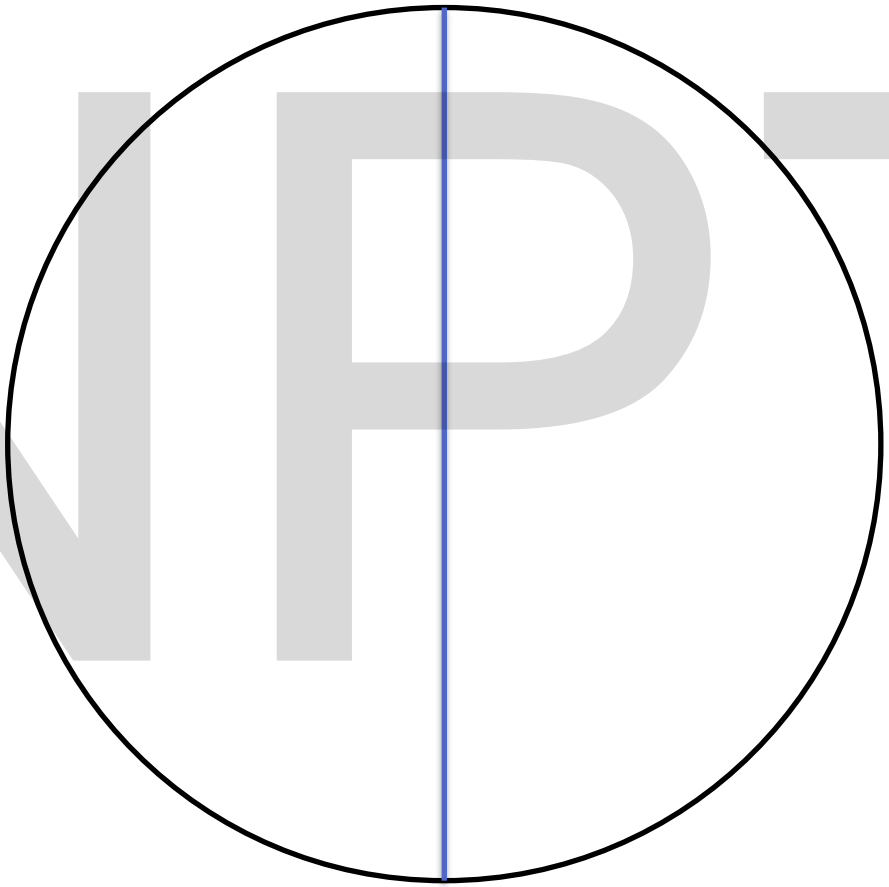
Size of K, V matrices

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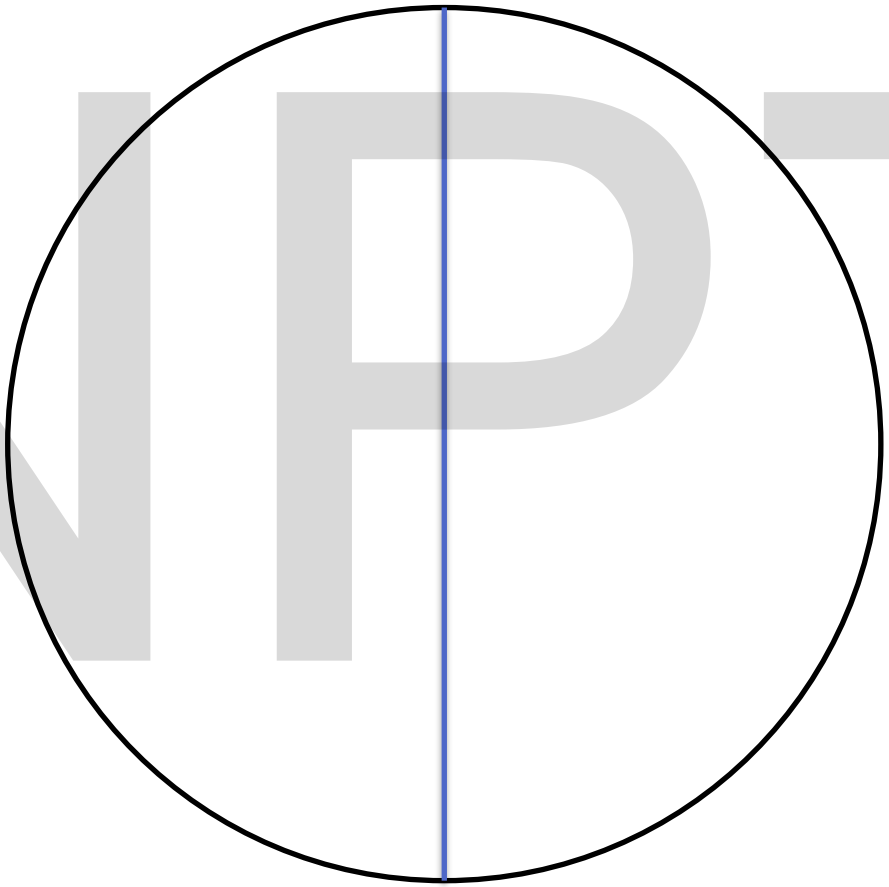
Exercise



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Exercise

What is the max batch size a Llama2-7B model can fit on a A100 GPU?

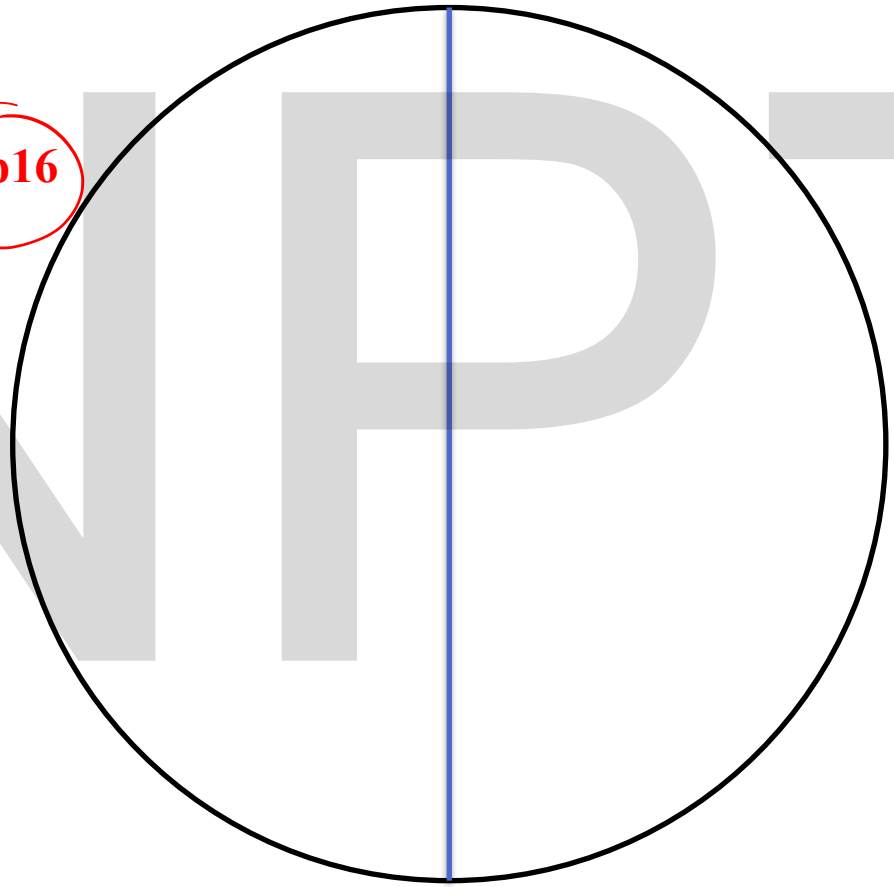


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What is the max batch size a Llama2-7B model can fit on a A100 GPU?

80GB
7B
Precision: fp16
Layers: 32
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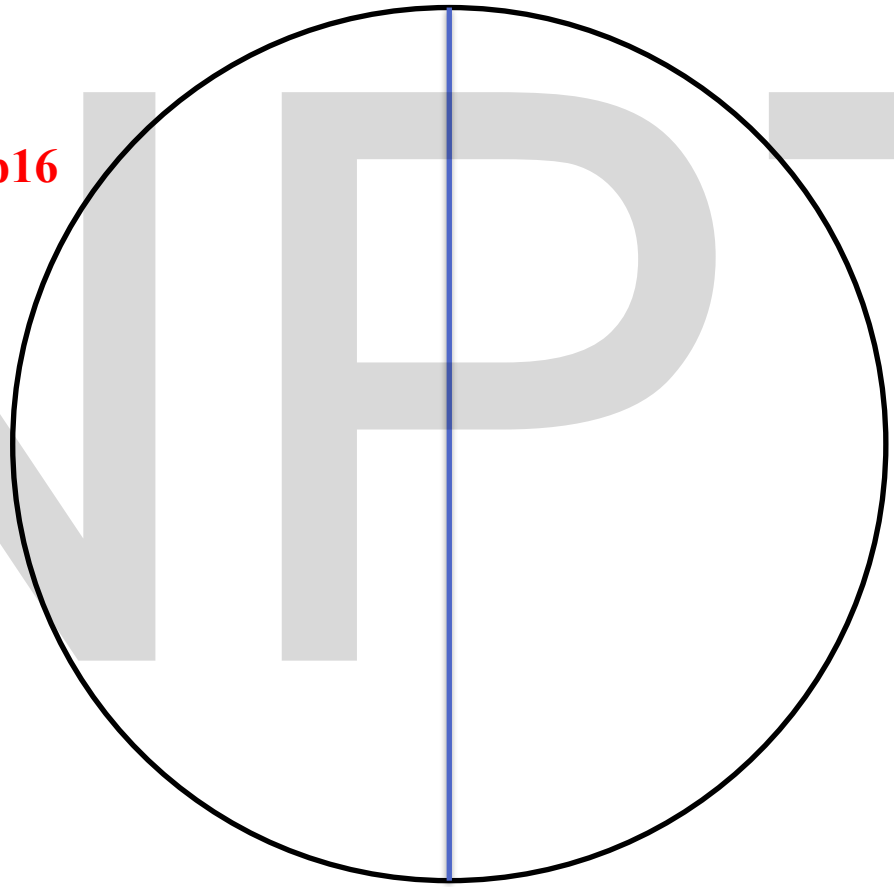


NPTEL

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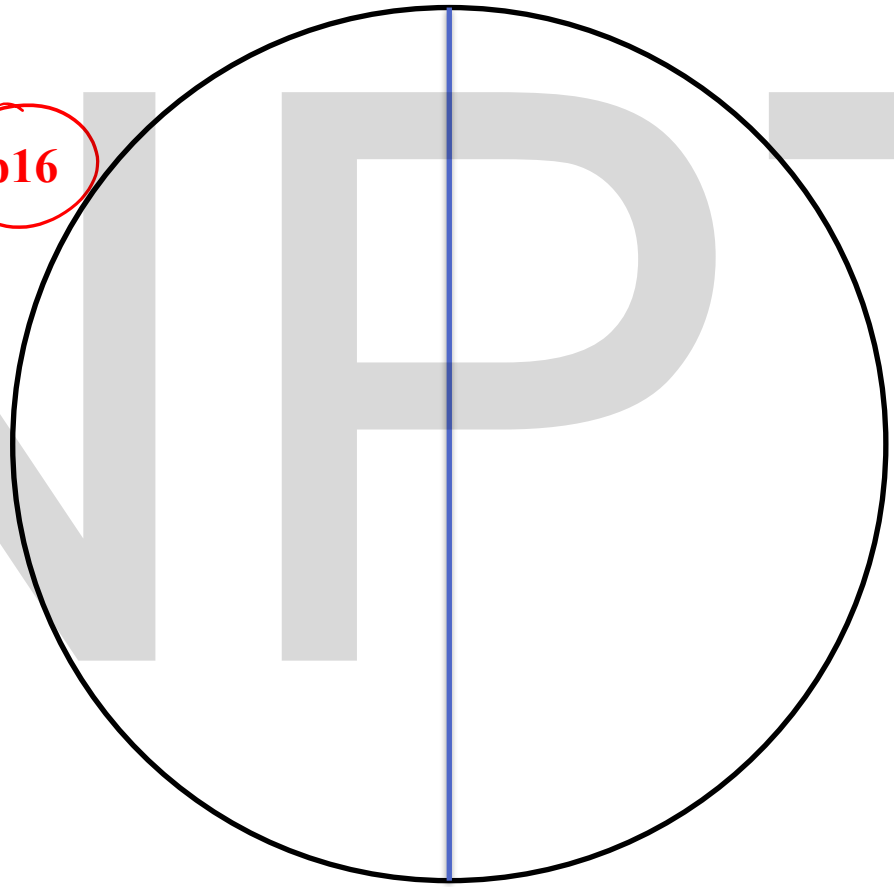
1. Model parameters :

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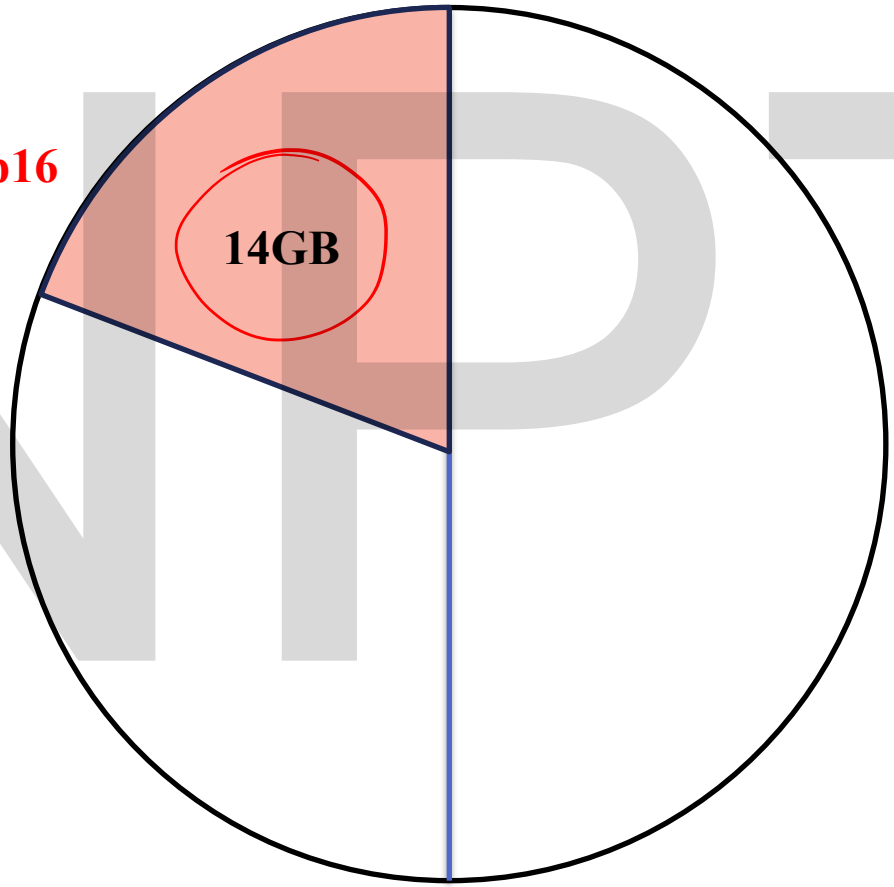
1. Model parameters : $Bytes_per_param * N_{parameters}$

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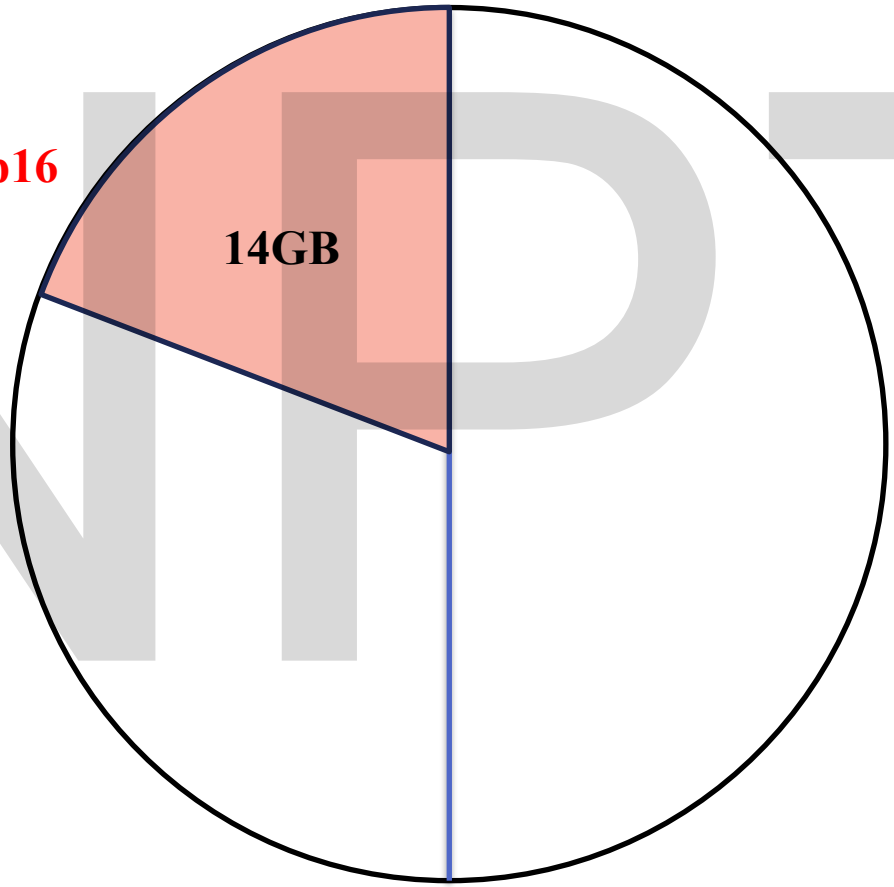
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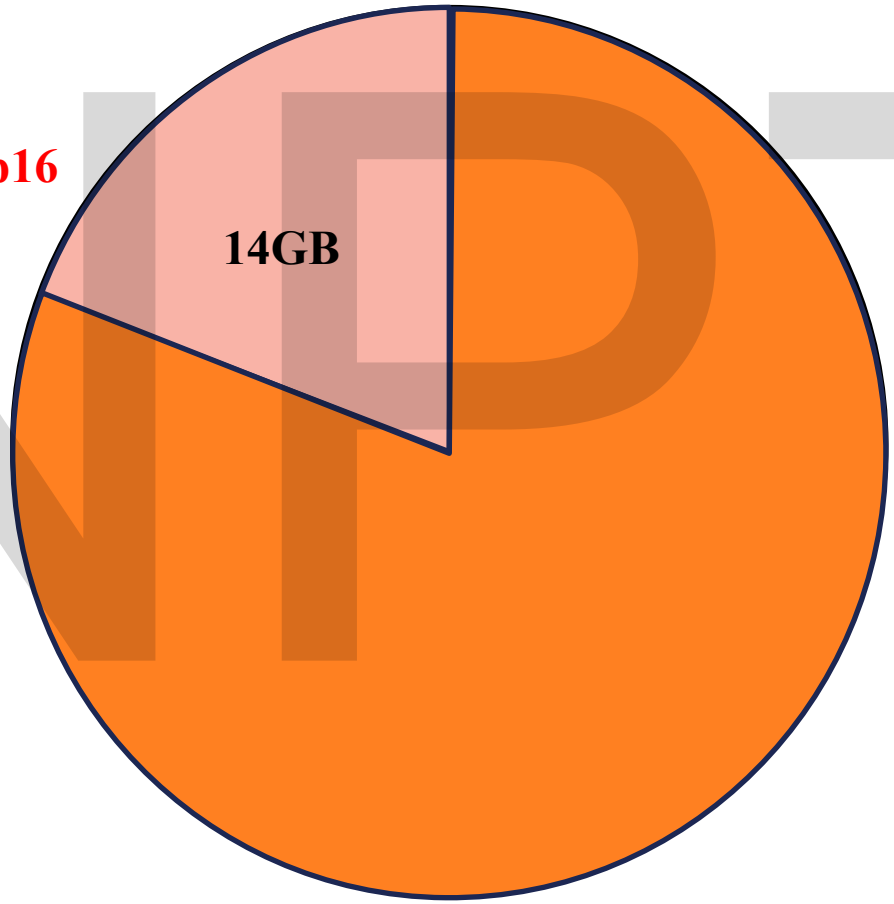
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2. KV cache per req : $2 * precision * N_{layers} * d_{model} * seqlen$

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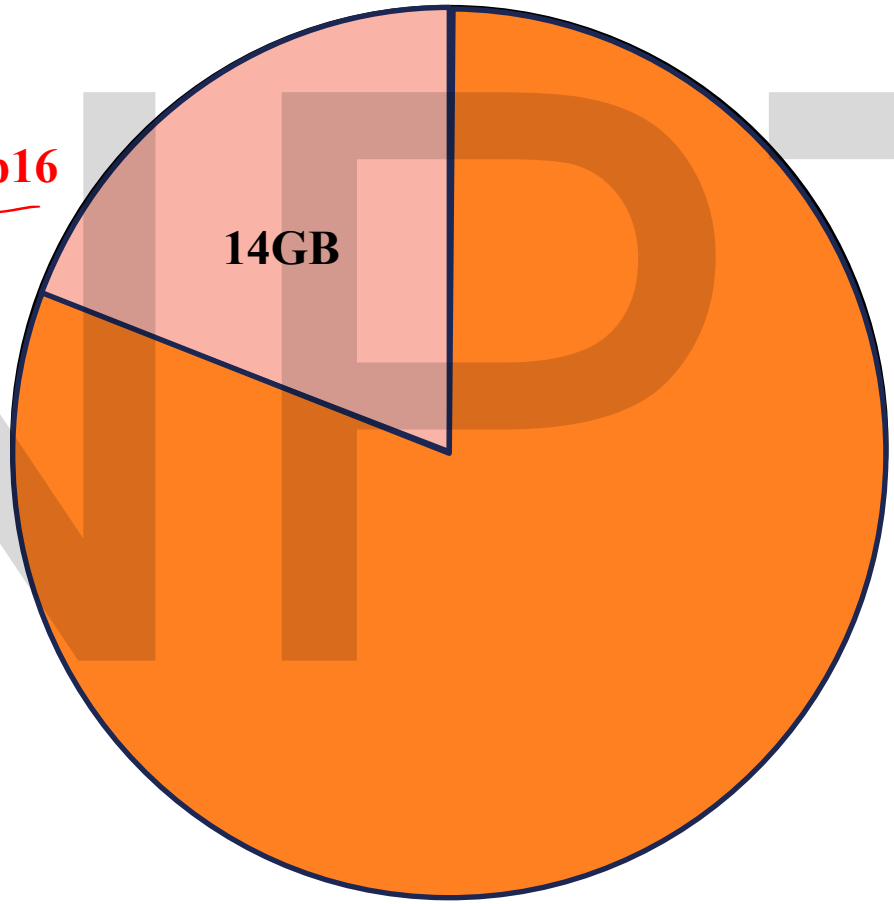
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1. Model parameters : $Bytes_per_param * N_{parameters}$

2. KV cache per req : $2 * precision * N_{layers} * d_{model} * seqlen$

$2*2*32*4096*4096 = 2GB$ per request

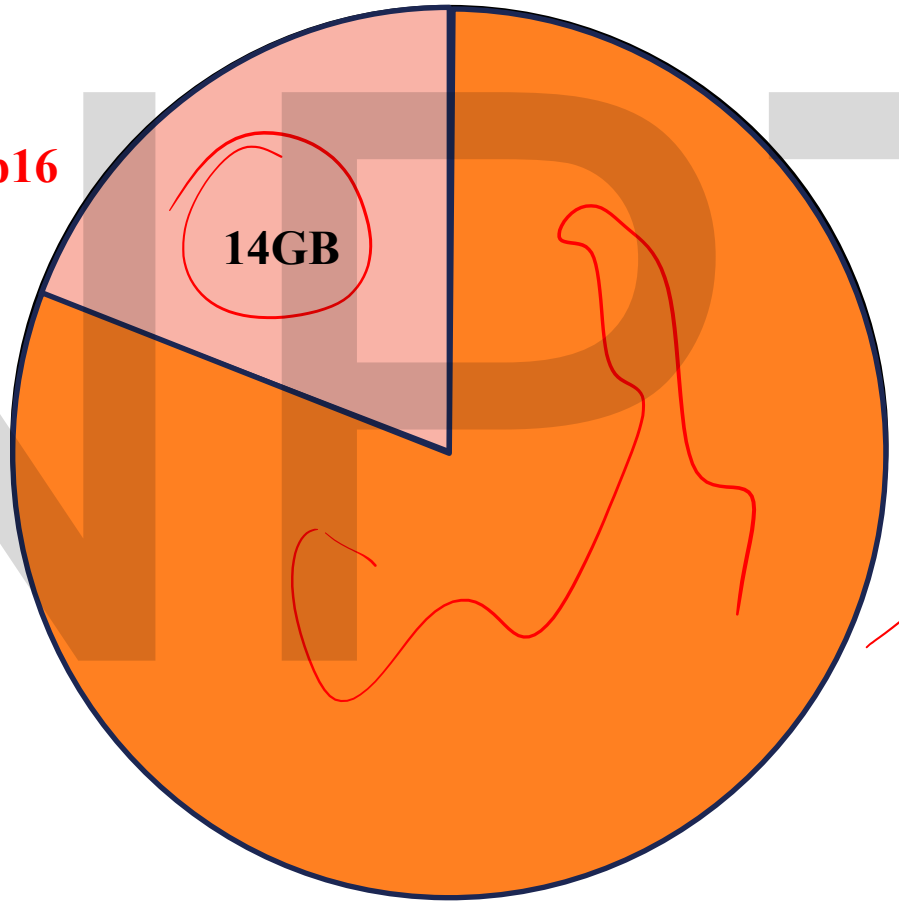
KV-cache

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1. Model parameters : $\text{Bytes_per_param} * N_{\text{parameters}}$

2. KV cache per req : $2 * precision * N_{layers} * d_{model} * seqlen$

$2*2*32*4096*4096 = 2\text{GB per request}$

Max batches = (GPU mem – Model param mem)/ KV_per_req

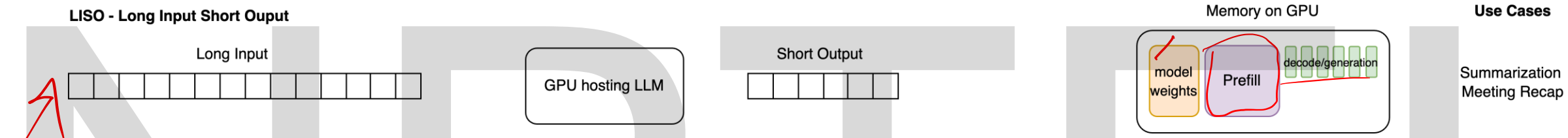
= (80-14)/2 = 33!

Agenda

- KV-Cache (contd.)
- **Paged-attention**

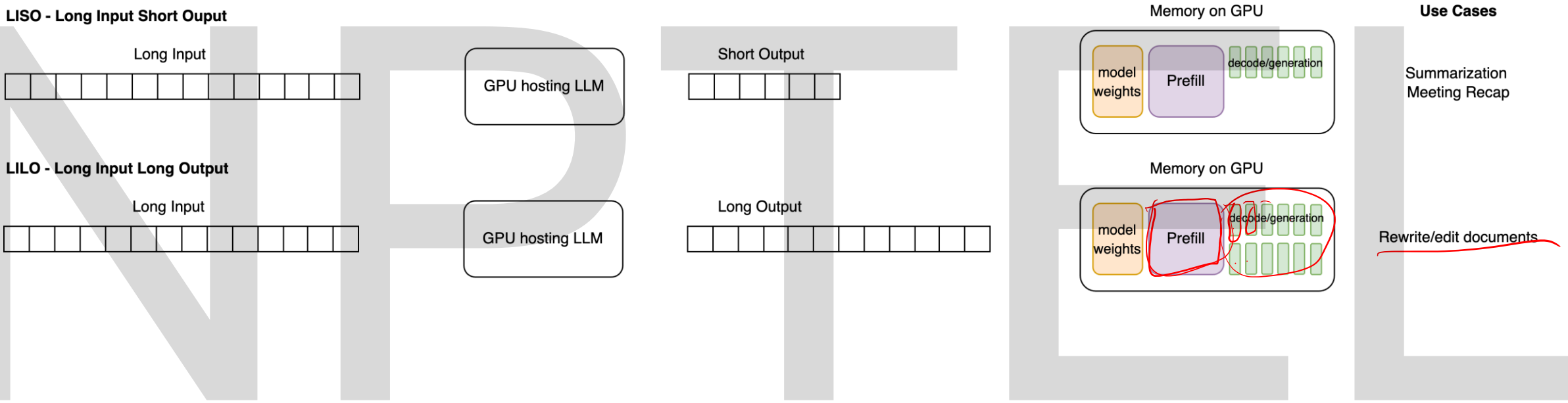
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Why KV cache memory management?

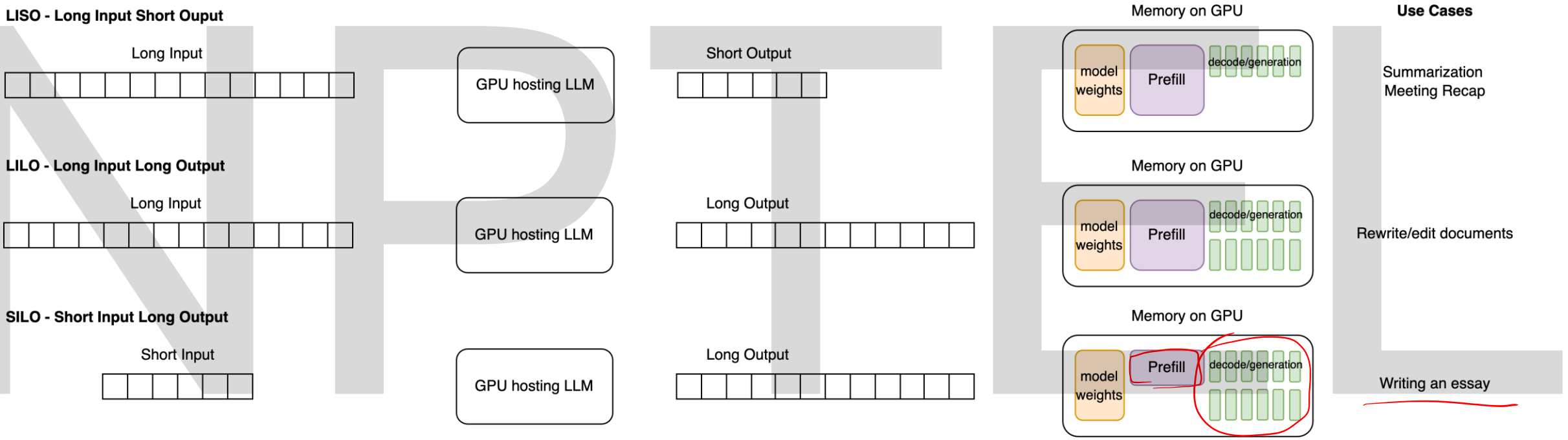


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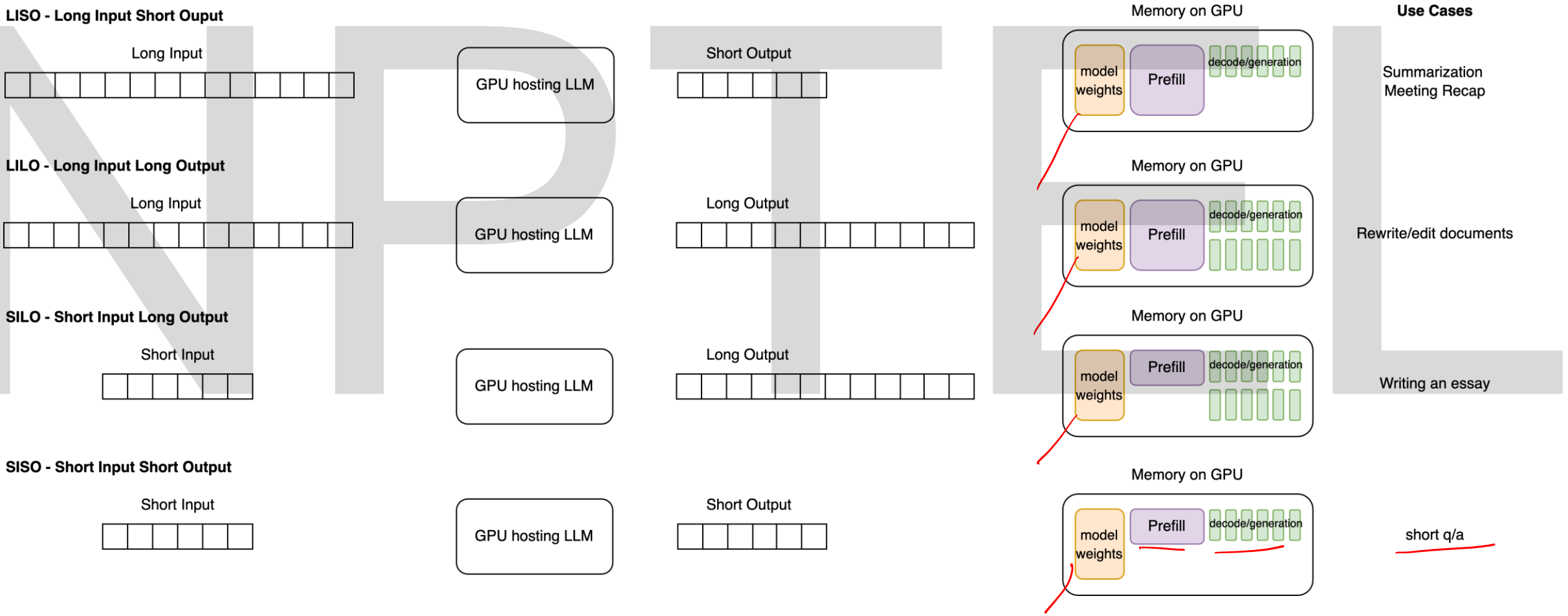
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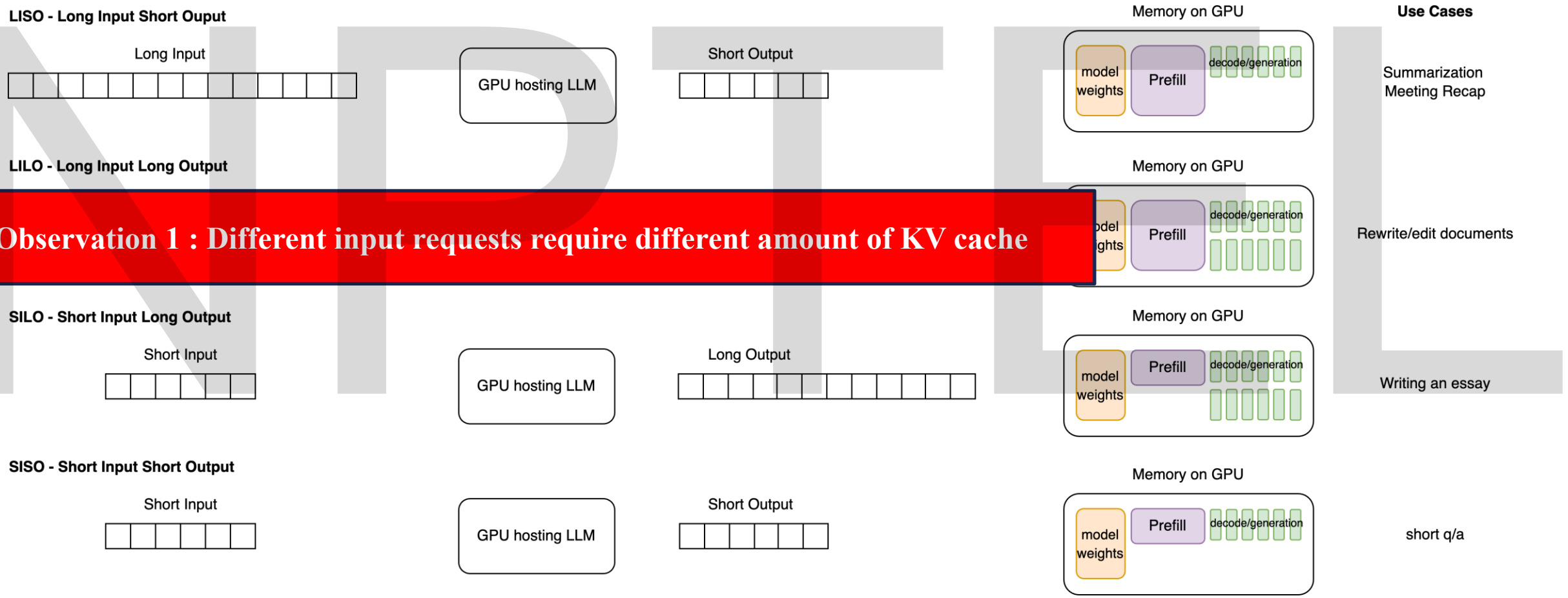
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Observation 2 : Tensor computations need contiguous memory allocation

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- Early systems like FasterTransformer allocate max model length for KV cache for each request

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Observation 2 : Tensor computations need contiguous memory allocation

- **Early systems like FasterTransformer allocate max model length for KV cache for each request**

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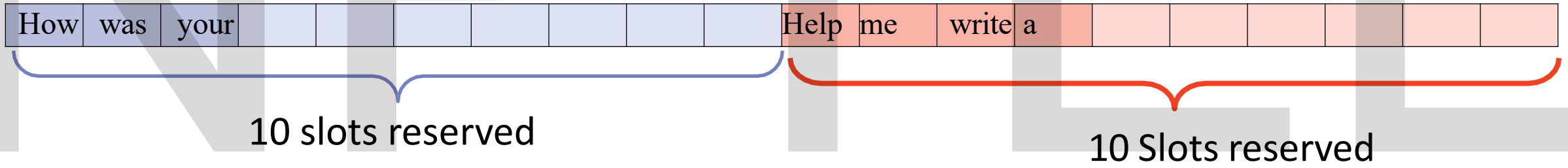
Prompt 1 :
“How was your”
Max model len : 10

Prompt 2 :
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Why KV cache memory management?

Observation 2 : Tensor computations need contiguous memory allocation

- **Early systems like FasterTransformer allocate max model length for KV cache for each request**

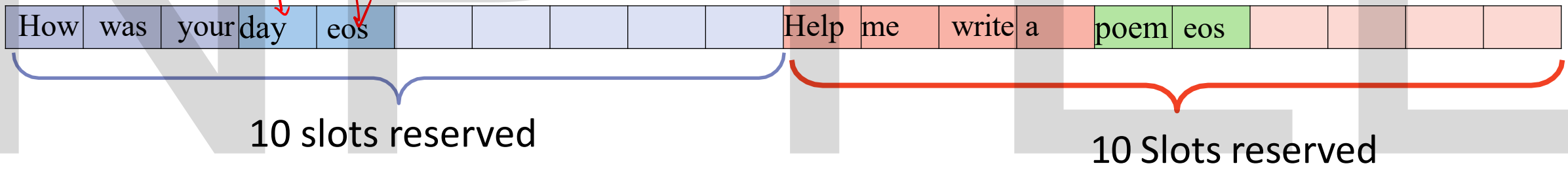


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Why KV cache memory management?

- **Early systems like FasterTransformer allocate max model length for KV cache for each request**



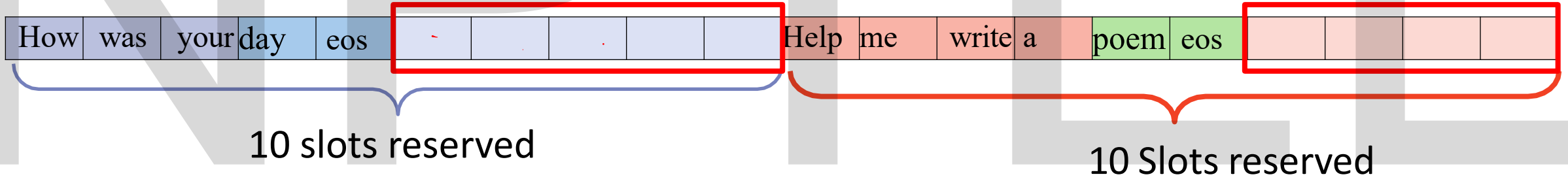
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Why KV cache memory management?

- **Early systems like FasterTransformer allocate max model length for KV cache for each request**



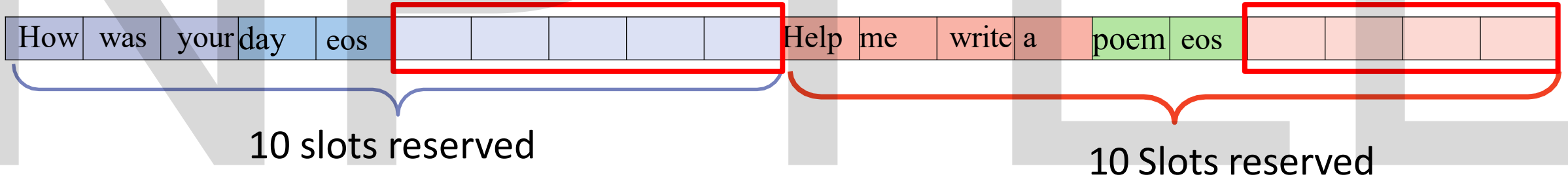
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Why KV cache memory management?

Observation 3 : Fixed preallocation results in internal fragmentation

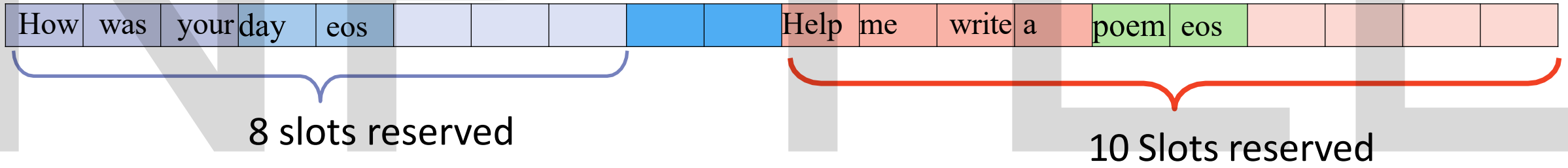
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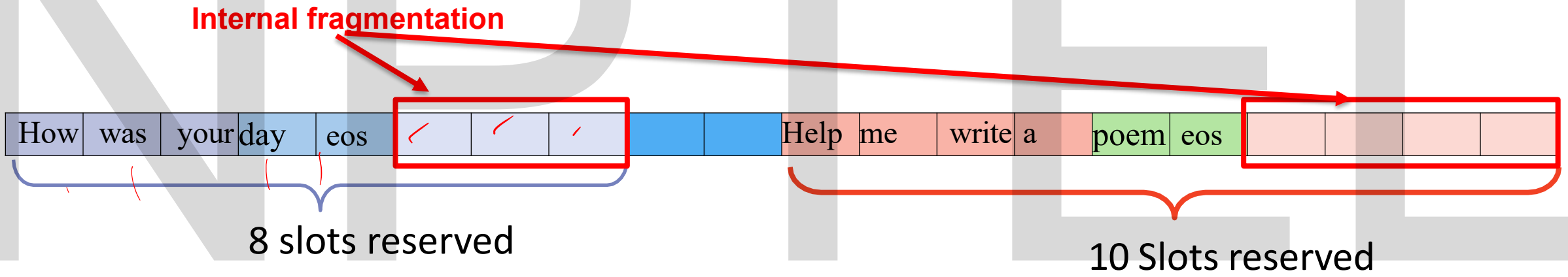
Why KV cache memory management?



Prompt 1 :
"How was your"
Max model len : 10
Max tokens : 8

Prompt 2 :
"Help me write a"
Max model len : 10
Max tokens : 10

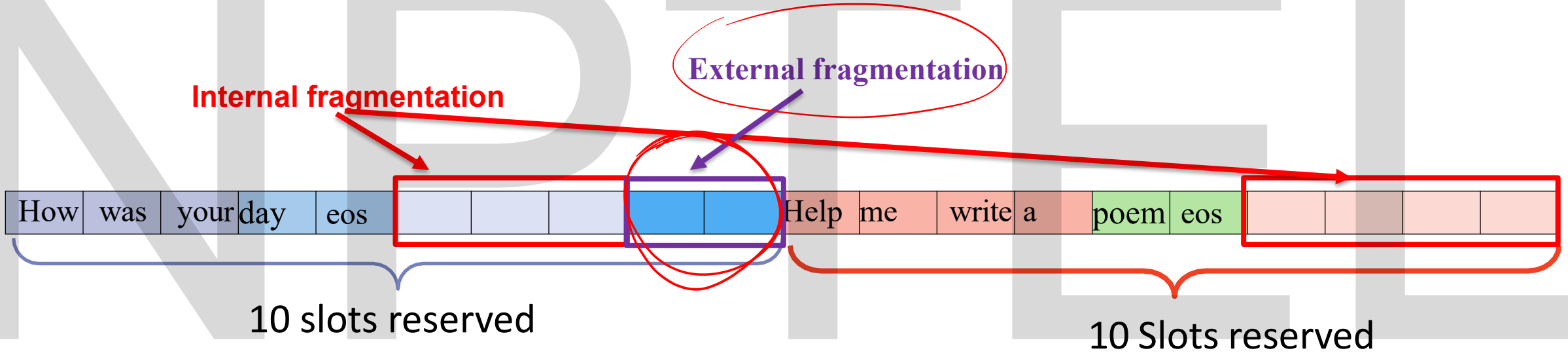
Why KV cache memory management?



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Why KV cache memory management?

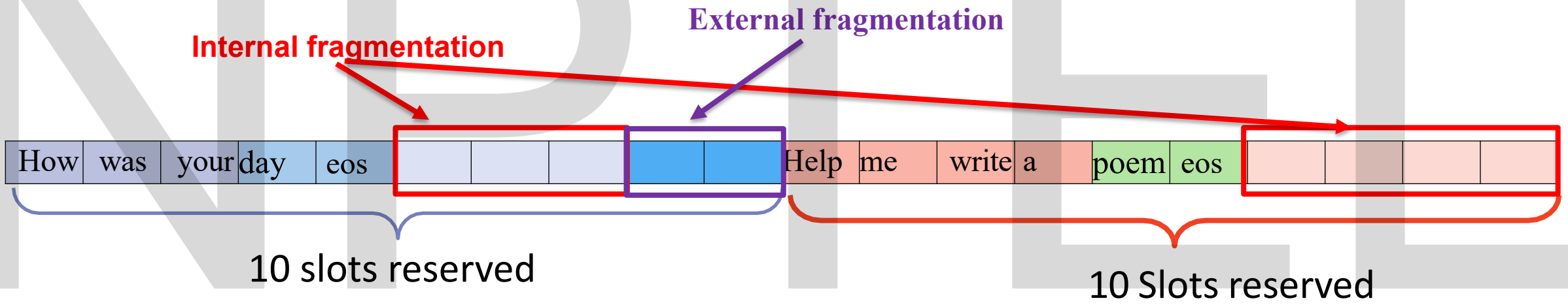


Prompt 1 :
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Why KV cache memory management?

Observation 4 : External fragmentation caused by buddy allocators



Prompt 1 :
"How was your"
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Prompt 2 :
"Help me write a"
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Max tokens : 10

Q: What if you could predict decode length exactly?

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Will that solve external and internal fragmentation?

NPTEL

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NPTEL

Q: What if you could predict decode length exactly?

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Is that perfect?

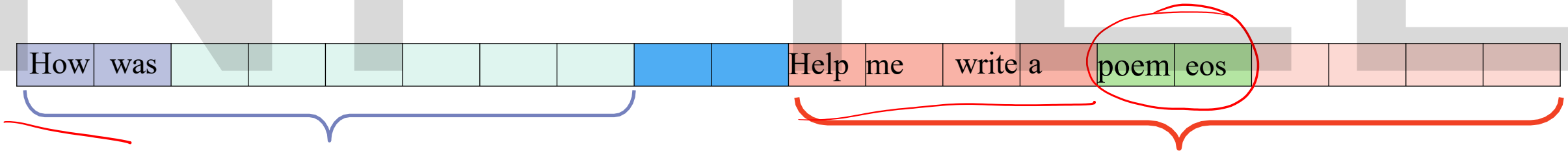
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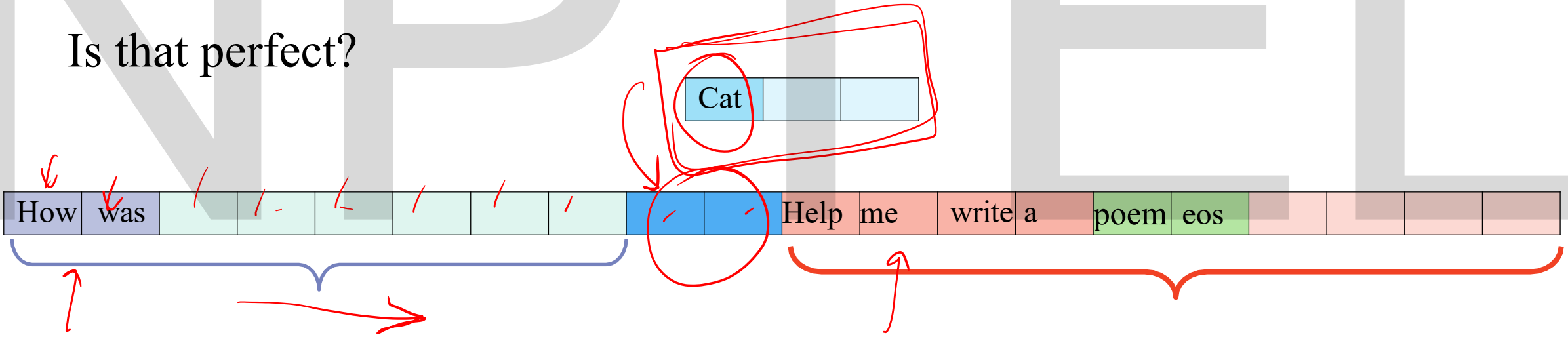
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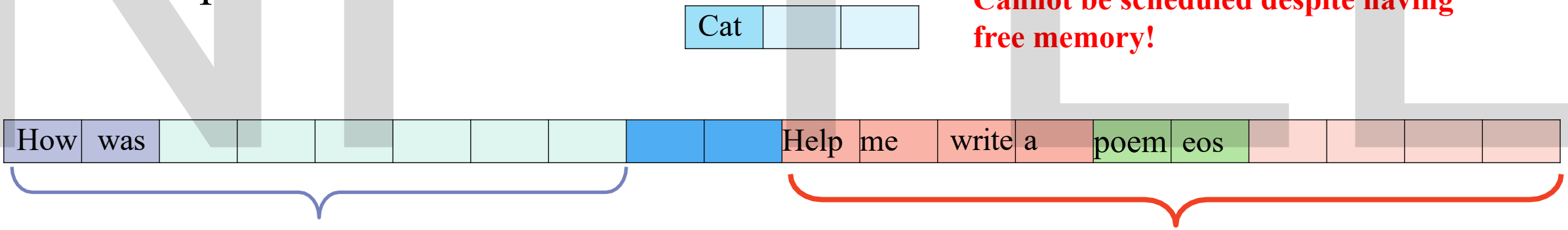
NPTEL

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Cannot be scheduled despite having free memory!

Two prompts with reserved spaces -> but the prompt grows one token (or block at a time)

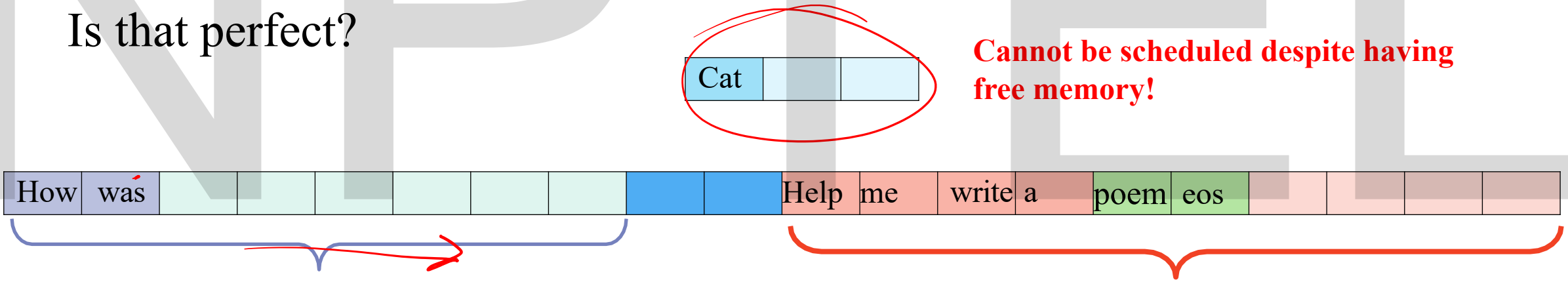
Meanwhile say we have a small prompt with just 3 tokens coming in

Q: What if you could predict decode length exactly?

Will that solve external and internal fragmentation?

Observation 5 : Memory needs grow one token at a time – reserved memory underutilized

Is that perfect?



Two prompts with reserved spaces -> but the prompt grows one token (or block at a time)

Meanwhile say we have a small prompt with just 3 tokens coming in

Chunked Preallocation - Drawbacks

- KV cache stored in contiguous memory
- Static allocation based on max request needs
- Ignores the fact that memory is needed incrementally and not all at once

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- KV cache stored in contiguous memory
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- **Internal Fragmentation due to overprovisioning for max tokens**

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Chunked Preallocation - Drawbacks

- KV cache stored in contiguous memory
- Static allocation based on max request needs
- Ignores the fact that memory is needed incrementally and not all at once

- **Internal Fragmentation due to overprovisioning for max tokens**
- **External Fragmentation due to buddy memory allocators**

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Chunked Preallocation - Drawbacks

- KV cache stored in contiguous memory
- Static allocation based on max request needs
- Ignores the fact that memory is needed incrementally and not all at once

- **Internal Fragmentation due to overprovisioning for max tokens**
- **External Fragmentation due to buddy memory allocators**
- **Underutilized reserved space due to static allocation**

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)